

# Low-Cost Multi-Modal Battery Diagnostic & Active Rebalancing System

Production Hardware Implementation, Calibration & Assembly Guide (BOM < \$45)

**SYSTEM HIGHLIGHTS:** Sub-microsecond acoustic Time-of-Flight (55 ps resolution with TDC7200), 16-bit electrical impedance sensing (INA226), multi-point thermal monitoring (NTC + MLX90614 IR), and closed-loop synchronous buck-boost active cell rebalancing with autonomous hardware lockout.

## 1. System Architecture & Measurement Principles

The diagnostic system interrogates battery cells using three physically complementary sensing modalities synchronized to an adaptive current excitation pulse:

- Ultrasonic Acoustic Sensing:** 1.0 MHz longitudinal compression waves propagate through the cell jelly-roll. Density and elasticity changes alter acoustic speed of sound (SOS) and attenuation. Gas generation drastically attenuates signal via acoustic mismatch ( $Z_{gas} \ll Z_{solid}$ ), while lithium plating creates density shifts with 180 deg phase inversion.
- Electrical Impedance Sensing:** High-frequency current pulses measure ohmic resistance ( $R_0$ ) and polarization impedance ( $R_1, C_1$ ) via a precision 4-wire Kelvin shunt and 16-bit Sigma-Delta ADC.
- Thermal Dynamics:** Surface thermistors and non-contact infrared sensors capture core-to-surface temperature gradients ( $dT/dt$ ) to identify localized Joule heating and internal micro-shorts.

## 2. Comprehensive Bill of Materials (BOM)

The complete hardware platform is designed for mass manufacturability with total unit cost below \$45:

| Item | Category       | Part Number / MPN          | Key Specs                                | Unit (\$) | Qty | Total (\$) |
|------|----------------|----------------------------|------------------------------------------|-----------|-----|------------|
| 1    | MCU            | ESP32-S3-WROOM-1           | Dual-core LX7 240MHz, 8MB Flash          | \$3.20    | 1   | \$3.20     |
| 2    | Ultrasonic TDC | TDC7200PWR (TI)            | 55 ps RMS resolution, SPI                | \$2.45    | 1   | \$2.45     |
| 3    | Ultrasonic LNA | AD8065ARTZ (ADI)           | 145 MHz FastFET OpAmp, low noise         | \$1.85    | 1   | \$1.85     |
| 4    | Comparator     | TLV3501 (TI)               | 4.5 ns ultrafast zero-crossing           | \$1.40    | 1   | \$1.40     |
| 5    | PZT Pulser     | TC4420EPA (Microchip)      | 6A peak gate driver (100ns pulse)        | \$1.25    | 1   | \$1.25     |
| 6    | Electrical AFE | INA226AIDGSR (TI)          | 16-bit I2C Current/Voltage/Power         | \$1.55    | 1   | \$1.55     |
| 7    | Thermal IR     | MLX90614ESF-BAA            | Non-contact IR, 0.02°C resolution        | \$5.80    | 1   | \$5.80     |
| 8    | NTC Sensors    | B57861S0103F040 (TDK)      | 10kΩ ±0.5%, Beta=3950                    | \$0.65    | 2   | \$1.30     |
| 9    | PZT Discs      | SMD10T2R111WL (PZT-5)      | 1.0 MHz, D=10mm, T=2mm                   | \$2.10    | 2   | \$4.20     |
| 10   | Power MOSFETs  | FDMS86180 (ON Semi)        | 100V, 60A, R <sub>ds</sub> = 3.2mΩ       | \$1.15    | 2   | \$2.30     |
| 11   | Buck Driver    | TPS28225 (TI)              | Synchronous Half-Bridge Driver           | \$1.10    | 1   | \$1.10     |
| 12   | Inductor       | SER2918H-103KL (Coilcraft) | 10μH, 14.5A I <sub>sat</sub> , 3.1mΩ DCR | \$3.40    | 1   | \$3.40     |
| 13   | Current Shunt  | CSS2H-2512R-L010F          | 10mΩ 0.1% 5W 4-terminal Kelvin           | \$1.15    | 1   | \$1.15     |

|               |              |                         |                                 |        |   |         |
|---------------|--------------|-------------------------|---------------------------------|--------|---|---------|
| 14            | Safety Relay | G3VM-61A1 (Omron)       | Solid State Opto-MOSFET Relay   | \$1.60 | 1 | \$1.60  |
| 15            | Passives/PCB | 4-Layer FR4 + Enclosure | 100x100mm 4-Layer ENIG Gold PCB | \$5.30 | 1 | \$5.30  |
| TO<br>TA<br>L |              |                         | Complete System Unit BOM Cost   |        |   | \$38.75 |

### 3. Schematics & Circuit Operation

#### 3.1 High-Voltage Ultrasonic Pulser & Analog Front-End (AFE)

The ultrasonic signal chain consists of an excitation pulser and a high-gain receiver front-end:

- Pulser:** The ESP32 RMT peripheral generates a 100 ns active-high gate pulse to the TC4420 driver. The driver switches an inductor-loaded boost circuit to produce a +48V spike into the 1.0 MHz PZT transmitting transducer (X1).
- LNA Receiver:** The acoustic echo received at X2 (~10-50 mV amplitude) is AC-coupled to the AD8065 JFET operational amplifier configured with a gain of +40 dB (100x) and a 2nd-order Sallen-Key bandpass filter centered at 1.0 MHz (Q=4).
- Zero-Crossing Timing:** The amplified echo feeds the TLV3501 ultrafast comparator. The comparator generates a clean digital STOP pulse into the TI TDC7200 Time-to-Digital Converter.
- Precision Timing Calculation:** The TDC7200 measures elapsed time between START (excitation trigger) and STOP with 55 ps resolution, transmitting the raw clock counts over SPI at 10 MHz.

#### 3.2 Synchronous Active Rebalancer Power Stage

The active rebalancing module employs a synchronous bidirectional buck-boost converter topology:

- **High-Efficiency Switches:** Two ON Semi FDMS86180 MOSFETs ( $R_{DS(on)} = 3.2\text{ m}\Omega$ ) driven by the TPS28225 synchronous driver with 25 ns internal adaptive dead-time.
- **Inductive Energy Transfer:** A Coilcraft 10  $\mu$ H, 14.5A saturation current inductor stores and shuttles charge between degraded cells and neighboring pack nodes with >92% energy efficiency.
- **Pulse Deplating Control:** When lithium plating is detected, the MCU initiates periodic micro-discharge pulses (15 ms pulse width, 1.0 Hz interval) to dissolve metallic lithium dendrites back into the electrolyte.

### 4. ESP32-S3 Pinout & Interface Matrix

| Pin Name   | GPIO         | Peripheral    | Connected Component | Signal Function                     |
|------------|--------------|---------------|---------------------|-------------------------------------|
| PULSE_TRIG | GPIO 4       | RMT CH0       | TC4420 Gate Driver  | 100 ns PZT Excitation Trigger       |
| TDC_START  | GPIO 5       | Output        | TDC7200 START Pin   | ToF Timing Start Reference          |
| TDC_INT    | GPIO 6       | Ext Interrupt | TDC7200 INTB Pin    | Measurement Complete ISR            |
| SPI_CS     | GPIO 10      | SPI2 CS       | TDC7200 CS Pin      | Chip Select (Active Low)            |
| SPI_SCK    | GPIO 12      | SPI2 SCK      | TDC7200 CLK Pin     | SPI Clock (10 MHz)                  |
| SPI_MISO   | GPIO 13      | SPI2 MISO     | TDC7200 DOUT Pin    | SPI Data Readout                    |
| SPI_MOSI   | GPIO 11      | SPI2 MOSI     | TDC7200 DIN Pin     | SPI Configuration Write             |
| I2C_SDA    | GPIO 21      | I2C0 SDA      | INA226 + MLX90614   | I2C Sensor Bus Data (400kHz)        |
| I2C_SCL    | GPIO 22      | I2C0 SCL      | INA226 + MLX90614   | I2C Sensor Bus Clock (400kHz)       |
| BAL_PWM    | GPIO 15      | LEDC PWM      | TPS28225 Driver     | 100 kHz Rebalancing PWM             |
| EMERG_TRIP | GPIO 16      | Output        | G3VM-61A1 Relay     | Hardware Pack Lockout Disconnect    |
| ADC_NTC1   | GPIO 1 (CH0) | ADC1          | NTC Thermistor #1   | Surface Temperature Sensing         |
| UART_TX/RX | GPIO 43/44   | UART0         | CP2102N USB Bridge  | Host Telemetry Stream (115200 baud) |

## 5. Transducer Mounting & Acoustic Coupling Guide

Acoustic impedance matching and stable contact pressure are paramount for sub-percent ToF diagnostic fidelity:

- **Acoustic Couplant Selection:** Use Dow Corning Sylgard 184 polydimethylsiloxane (PDMS) elastomer pads ( $Z \sim 1.5$  MRayl) for permanent vibration-resistant mounting, or medical ultrasound gel (Aquasonic 100) for benchtop testing.
- **Clamping Force:** Apply a calibrated normal force of  $7.5 \pm 1.5$  N per transducer using the spring-loaded 3D-printed PETG fixture. Excessive force deforms cylindrical 18650 casing walls; insufficient force introduces micro-air gaps ( $Z_{\text{air}} = 0.0004$  MRayl, causing 99.9% acoustic reflection).
- **Axial Alignment:** Position Tx and Rx transducers strictly along the diametric centerline ( $180^\circ \pm 2^\circ$  opposition) at the vertical midpoint ( $Z = 32.5$  mm) to avoid casing end-cap reflections.

## 6. Firmware Flashing & Zero-Point Calibration

### 6.1 PlatformIO / ESP-IDF Build & Flash:

```
# 1. Clone repository and navigate to firmware directory
cd firmware

# 2. Build firmware image with high-frequency FreeRTOS timers
pio run -e esp32-s3

# 3. Flash to ESP32-S3 over USB CDC COM port
pio run -e esp32-s3 -t upload --upload-port COM3

# 4. Open serial telemetry monitor (115200 baud, 8N1)
pio device monitor -b 115200
```

### 6.2 Two-Point Calibration Protocol:

1. **Zero-Point ToF Calibration:** Place transducers in direct face-to-face contact with couplant ( $d=0$  mm). Record intrinsic electronic delay  $t_{\text{prop}} \sim 120$  ns.
2. **Baseline Reference Calibration:** Mount transducers on a known healthy baseline cell (18650, SOC=50%,  $T=25.0^\circ\text{C}$ ). Verify measured ToF matches canonical standard ( $t_{\text{ToF}} = 7.20 \mu\text{s} \pm 0.15 \mu\text{s}$ ).
3. **Electrical Shunt Calibration:** Apply a precision 1.000A constant current load and calibrate the INA226 calibration register (CAL = 0x0A00).

## 7. Multi-Layer Hardware & Software Safety Interlocks

The hardware architecture enforces a 3-tier fail-safe safety interlock hierarchy:

- **Tier 1 (Autonomous Hardware Trip < 50  $\mu\text{s}$ ):** Comparator-based overcurrent lockout shuts down gate driver TPS28225 PWM if current exceeds 6.5A or temperature exceeds  $65^\circ\text{C}$ , de-energizing the opto-isolated solid state relay (K1).
- **Tier 2 (Firmware Safety Supervisor < 5 ms):** Core 1 FreeRTOS high-priority task checks telemetry limits ( $V_{\text{min}}=2.5\text{V}$ ,  $V_{\text{max}}=4.25\text{V}$ ,  $T_{\text{max}}=55^\circ\text{C}$ ). Detects voltage collapse or thermal runaway triggers instantaneous cell isolation.
- **Tier 3 (ML Multi-Modal Reasoning Interlock < 50 ms):** ML processor classification of internal\_short triggers CRITICAL\_LOCKOUT\_ISOLATED state in the active rebalancing decision engine.

## 8. Step-by-Step Benchtop Validation Checklist

| Step | Validation Phase          | Procedure & Acceptance Criteria                                                         | Status                        |
|------|---------------------------|-----------------------------------------------------------------------------------------|-------------------------------|
| 1    | Power Rails Check         | Verify 3.3V, 12V, 48V boost rails within $\pm 2\%$ tolerance with DMM.                  | <input type="checkbox"/> PASS |
| 2    | SPI & I2C Bus Discovery   | Verify TDC7200 (SPI) and INA226 (I2C addr 0x40) respond with valid Device IDs.          | <input type="checkbox"/> PASS |
| 3    | PZT Pulser Ringing Check  | Inspect Tx 50V pulse on oscilloscope. Confirm rise time < 25 ns and damping < 3 cycles. | <input type="checkbox"/> PASS |
| 4    | LNA Signal-to-Noise Ratio | Observe received echo on oscilloscope. Confirm SNR > 26 dB and peak amplitude > 1.2V.   | <input type="checkbox"/> PASS |
| 5    | TDC Timing Stability      | Record 100 consecutive ToF samples. Standard deviation must be < 500 ps.                | <input type="checkbox"/> PASS |
| 6    | Rebalancing Power Stage   | Test bidirectional current transfer at 0.5A, 1.0A, 2.5A. Verify efficiency > 90%.       | <input type="checkbox"/> PASS |
| 7    | Emergency Lockout Test    | Simulate internal short fault signal. Confirm relay disconnects within 50 $\mu$ s.      | <input type="checkbox"/> PASS |
| 8    | End-to-End Live Stream    | Connect to FastAPI backend over USB UART. Confirm 10Hz DiagnosticFrame streaming.       | <input type="checkbox"/> PASS |